

CS 270: DIGITAL IMAGE PROCESSING (SPRING 2024)

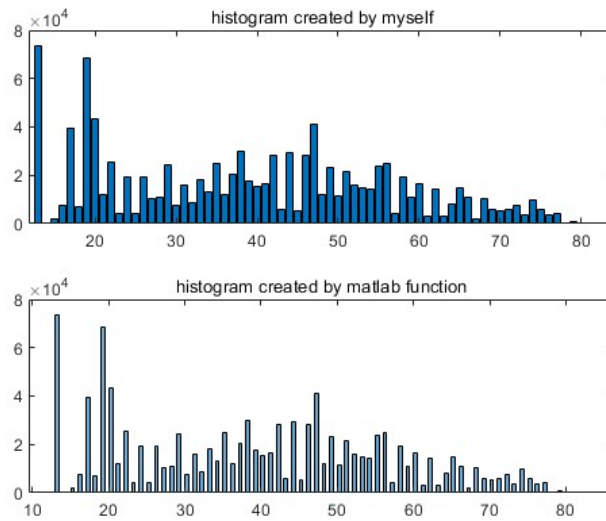
Assignment 1

Due: 23:59, April 7, 2024

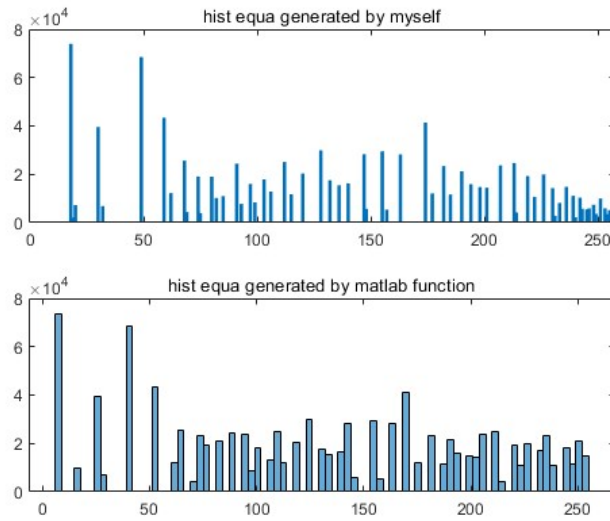
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Problem 1: Histogram Equalization

- (a) Compute the histogram of `grain.tif` and show the histogram image in your report. (Built-in functions `hist` and `histogram` are not allowed). (5 points)



- (b) Implement histogram equalization on `grain.tif`. Show the histogram equalized image and the histogram of equalized image in your report. (Built-in function `histeq` is not allowed). (10 points)



- (c) Implement contrast limited adaptive histogram equalization (CLAHE) on `tire.tif`, where you traverse every pixel with a 41×41 patch and process histogram equalization within each patch and update the center patch of 3×3 . Each time you move the patch with a step of 1 pixel. The clip limit for CLAHE is 0.02, which means that after the normalization of the histogram, amplification above 0.02 should be clipped and evenly distributed to other parts of the histogram. Show the CLAHE processed images and corresponding histogram in your report. (Built-in function `adapthisteq` is not allowed) (20 points)

original image



hist equalization img



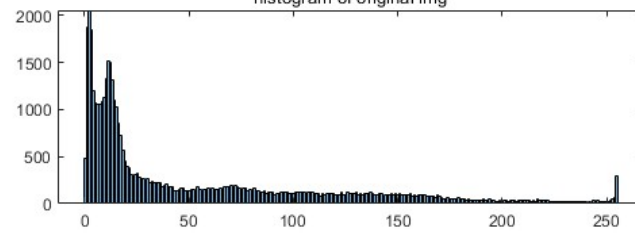
CLAHE by matlab



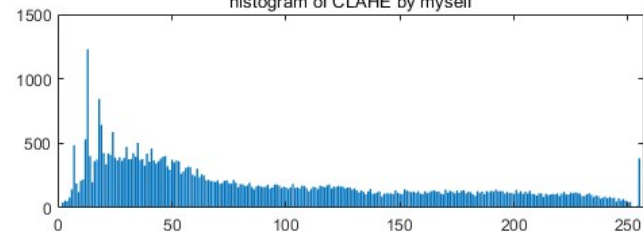
CLAHE by myself



histogram of original img



histogram of CLAHE by myself



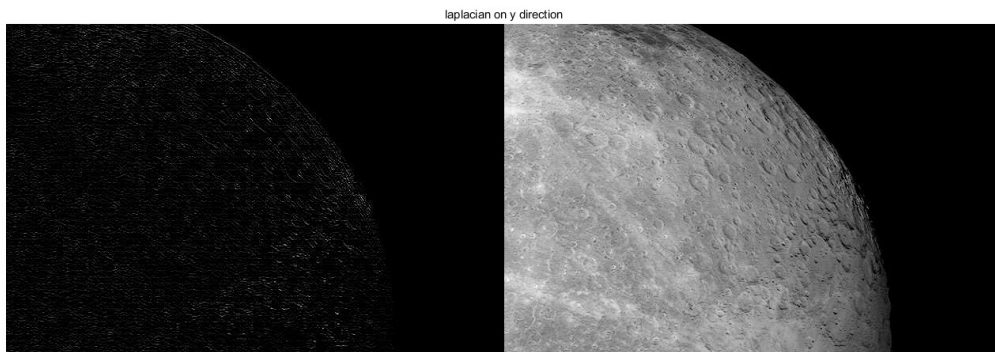
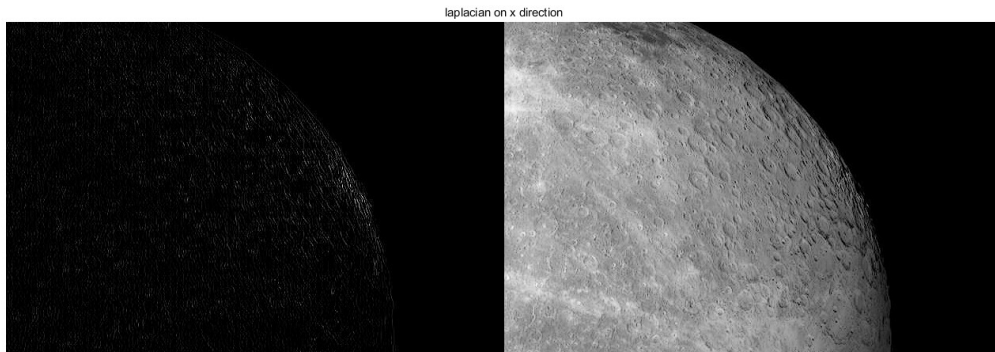
Problem 2: Image Sharpening

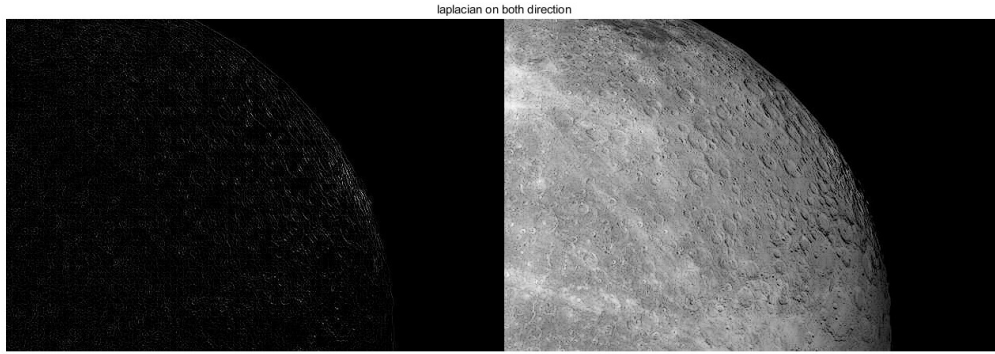
(a) Apply the 3×3 Laplacian kernels (x -direction and y -direction) on `moon.jpg` to obtain the details of the image. Since the Laplacian kernels are separable, please separate them into combinations of 1-D kernels and then apply them to the image sequentially. Show the separated Laplacian kernels and the corresponding processed images. (Implement your own **convolution** operator.) (15 points)

Applying Laplacian kernel,

on x direction: $\begin{bmatrix} 1 & -2 & 1 \end{bmatrix}$

on y direction: $\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$

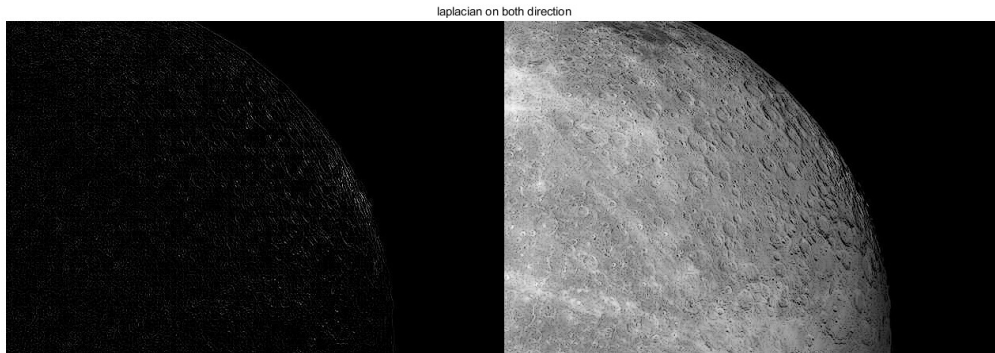




(b) Apply the 3×3 Laplacian kernels (not separated) to sharpen the image by 2-D convolution. Show the results in your report. (10 points)

Applying Laplacian kernel not seperated,

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$



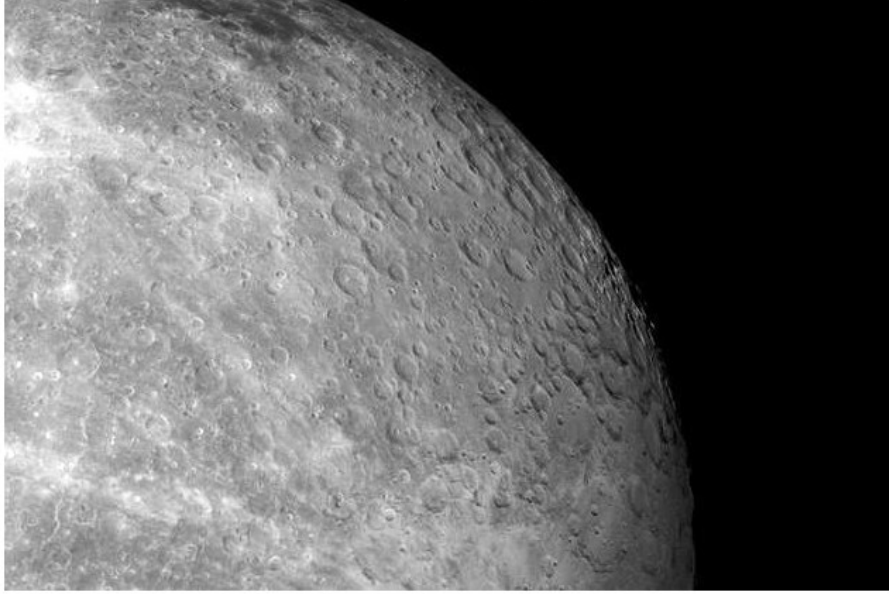
(c) Use unsharpen mask to sharpen the image. Show the results in your report. (10 points)

using unsharpen mask,

$$g_{mask}(x, y) = f(x, y) - f(\bar{x}, y),$$

$$g(x, y) = f(x, y) + k * g_{mask}(x, y), k = 1$$

unsharpen mask image



Problem 3: Nonlinear Filter

Apply the median filter and Gaussian filter to the image `lena_noisy.tif`. In your report, Please show the images processed by the median filter and Gaussian filter respectively, and analyze the cause of the results. (Filtering operations cannot be implemented by calling functions, Hint: Choose the best kernel size you think, $\sigma = 1$) (30 points)

median filter can filter salt-and-pepper noise better than gaussian filter

the larger the kernel size is , the less noise, but the more blurry the image is

image with median filter kernel size=3(left) kernel size=5(right)



image with gaussian filter kernel size=3(left) kernel size=7(right)

